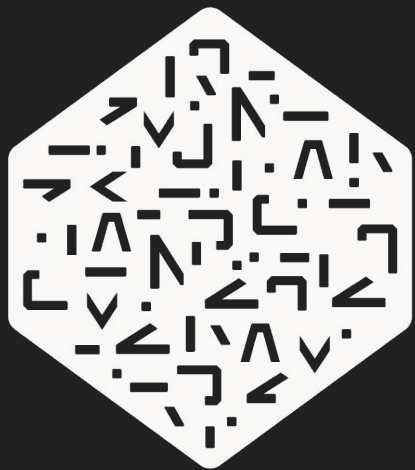




NUMERAI

MikeP

Legal Disclaimer

- This presentation is not to be viewed as an advertisement or a solicitation to invest in Numerai or any of its pooled investment vehicles.
- Presentation cannot be distributed with express written consent of Numerai.
- This presentation shall not be construed as investment advice.

<https://numer.ai/terms>

Getting started on Numerai

```
> pip install numerapi

from numerapi import NumerAPI
api = NumerAPI()
api.download_dataset(
    "v5.2/train.parquet",
    "train.parquet"
)
api.download_dataset(
    "v5.2/validation.parquet",
    "validation.parquet"
)
api.download_dataset(
    "v5.2/validation_example_preds.parquet",
    "validation_example_preds.parquet"
)
api.download_dataset(
    "v5.2/live.parquet",
    "live.parquet"
)
api.download_dataset(
    "v5.2/live_example_preds.parquet",
    "live_example_preds.parquet"
```



Tournaments ▾ Hedge Fi

NUMERAI TOURNAMENT

Overview

Data

Models

Submissions

Scoring

Staking

Bounties

NUMERAI SIGNALS

Overview

Data

Models

Submissions

Scoring

Staking

Signals + QuantConnect

NUMERAI TOURNAMENT

Overview

Everything you need to know to get started in under 5 minutes!

Introduction

Numerai is a data science competition where you build machine learning models on financial data. You are provided with free, high quality data that you can use to train models. Numerai computes the performance of these predictions over the following period, to earn (or burn) based on your model's performance.

You can [sign up](#) and visit your [home page](#) for a full suite of tutorials.

Data

Numerai's free dataset is made of clean and regularized financial data. The data can be given out for free and modeled without any financial domain knowledge. Models you build on this data cannot be used outside of the Numerai tournament.

Obsolete

```
> pip install numerapi

from numerapi import NumerAPI
api = NumerAPI()
api.download_dataset(
    "v5.2/train.parquet",
    "train.parquet"
)
api.download_dataset(
    "v5.2/validation.parquet",
    "validation.parquet"
)
api.download_dataset(
    "v5.2/validation_example_preds.parquet",
    "validation_example_preds.parquet"
)
api.download_dataset(
    "v5.2/live.parquet",
    "live.parquet"
)
api.download_dataset(
    "v5.2/live_example_preds.parquet",
    "live_example_preds.parquet"
```



NUMERAI TOURNAMENT

Overview

Data

Models

Submissions

Scoring

Staking

Bounties

NUMERAI SIGNALS

Overview

Data

Models

Submissions

Scoring

Staking

Signals + QuantConnect

NUMERAI TOURNAMENT

Overview

Everything you need to know to get started in under 5 minutes!

Introduction

Numerai is a data science competition where you build machine learning models on a financial market. You are provided with free, high quality data that you can use to train models daily. Numerai computes the performance of these predictions over the following NMP using your model, to earn (or burn) based on your model's performance.

You can [sign up](#) and visit your [home page](#) for a full suite of tutorials.

Data

Numerai's free dataset is made of clean and regularized financial data. The data can be given out for free and modeled without any financial domain knowledge. Models you build on this data cannot be used outside of the Numerai tournament.

The AI era is upon us

How many r's in are in strawberry



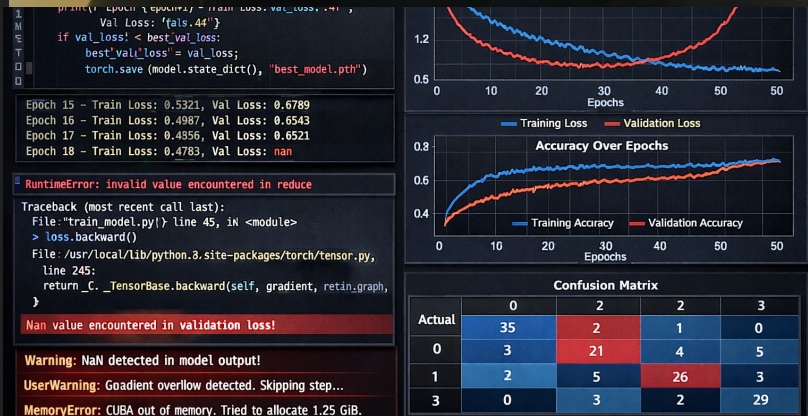
Thought for a couple of seconds

There are **three** "r"s in the word "strawberry."



It is easier than ever to play Numerai

2025



2026



ChatGPT 5.2 Thinking ▾

Make me a good numerai model

How to develop with AI in 2026

Teach AI to fish

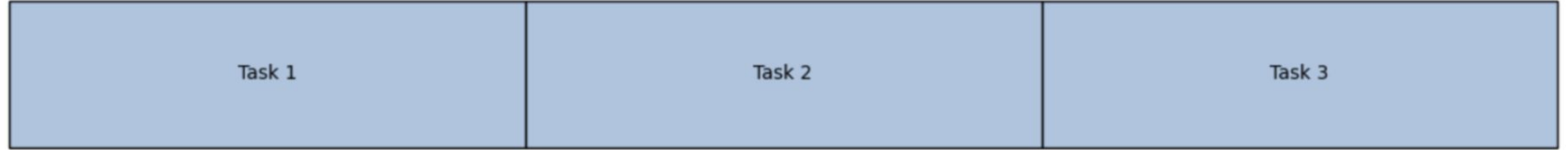
Baseline: Each task long time

No AI

Task 1

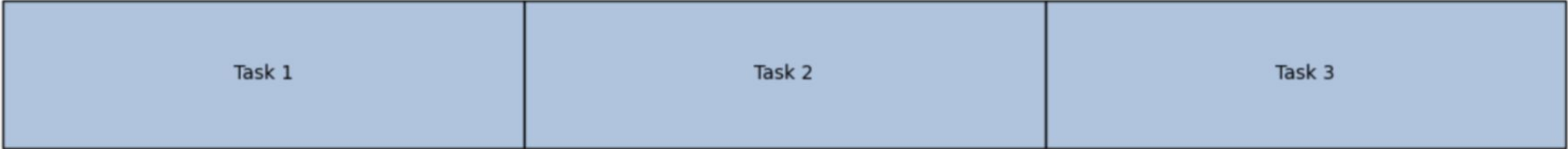
Task 2

Task 3



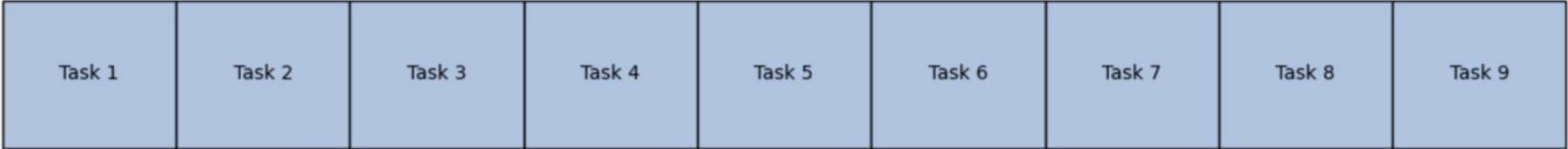
Baseline: Each task long time

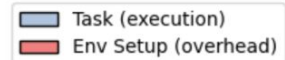
No AI



With AI Assistance: Much faster tasks

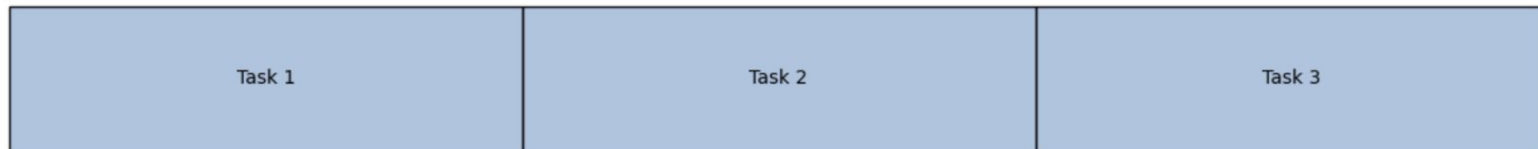
AI-assisted





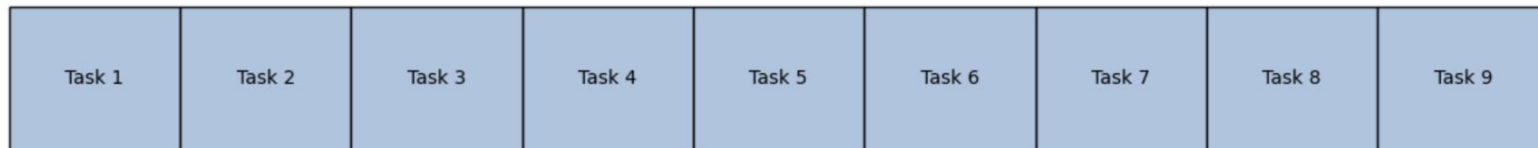
Baseline: Each task long time

No AI



With AI Assistance: Much faster tasks

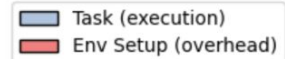
AI-assisted



Set up env so that AI can do each task in full

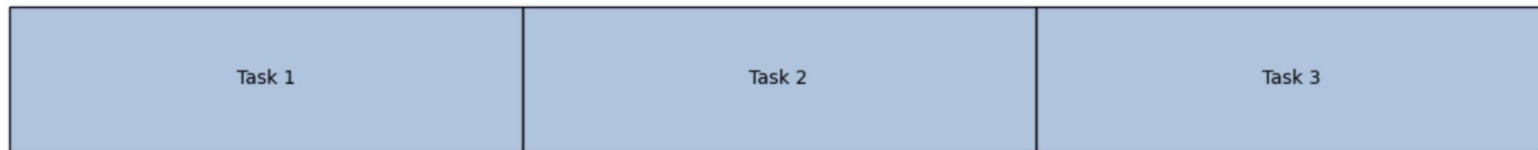
AI Automation





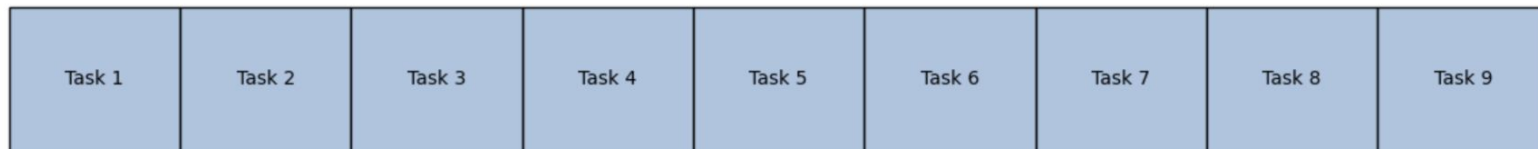
Baseline: Each task long time

No AI



With AI Assistance: Much faster tasks

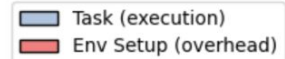
AI-assisted



Set up env so that AI can do each task in full
Setup overhead decreases as the environment improves

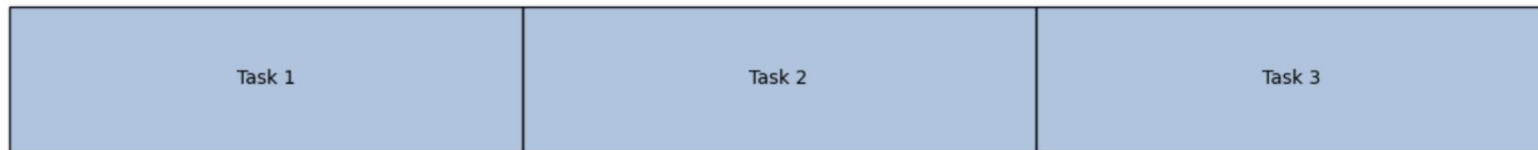
AI Automation





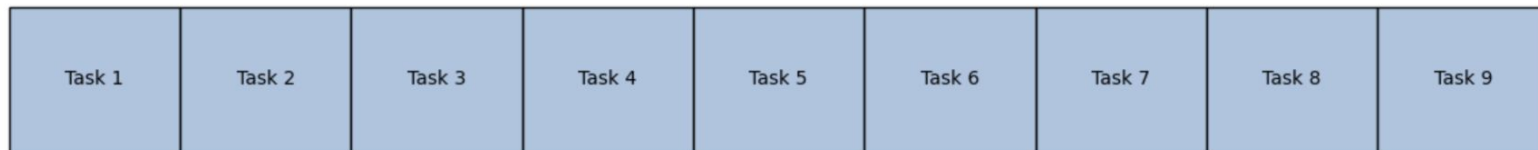
Baseline: Each task long time

No AI



With AI Assistance: Much faster tasks

AI-assisted



Set up env so that AI can do each task in full
Setup overhead decreases as the environment improves

AI Automation



What does it mean to improve "Env Setup"?

AI Driven Development 101

Get out of the way

1. Give AI access
2. Give AI context
3. Teach AI the specifics

Basic AI coding stack we use at Numerai

1. Give AI access - Codex CLI / Claude Code CLI
2. Give AI context - [AGENTS.md](#)
3. Teach AI the specifics - SKILL.md

CLI - keeps you from breaking the AI loop

Why CLI?

- CLI can already see your code (no copy and pasting)
- Already has git access and infra access to do auxiliary necessary tasks
- Has correct python env for actually running code

CLI example

1. `brew install codex`
2. `cd path/to/project`
3. `codex`

› user said there's a bug where tooltip doesn't show on signals leaderboard. Plz fix

- **Plan**
 - └ Fixing Signals leaderboard tooltip bug
 - Locate Signals leaderboard UI code
 - Identify and fix issue
 - test and validate
 - open PR and inform user of task completion

Give the AI context

Without Context

You: Go evaluate a Meta Model that only uses models built on the latest dataset

...

I will use a standard ridge regression to blend the benchmark models together into a "Meta Model"

"Just prompt better"

Go evaluate a Meta Model which only uses v5.2 models.

The Meta Model generation code can be found in
path/to/meta_model/code

You can determine which models are v5.2 by running the proper query from the tournament db. Details about that can be found in path/to/tournament/documentation

By default we use SWMM for blending (stands for Stake-Weighted)....

Test any new code written before you come back to me...

AGENTS.md

1. Explain the context of the folder / repo
2. Give high level organization overview
3. Give general development guidance

AGENTS.md

Monorepo Guide

This repository combines many projects in one place for the company Numerai.

General layout

Numerai is a quantitative hedge fund which gives out anonymized data each day for a data science tournament.

It then receives predictions back from the data science tournament, and combines those into a Meta Model.

The tournament code itself is handled by a different repo, but the Meta Model is built in `monorepo/data_generation` orchestrated by `monorepo/airflow`.

Then `monorepo/trading` will read the Meta Model alpha scores and run the portfolio optimizer and tradelist generation code in order to facilitate the hedge fund's actual trading.

Dev Guide

- Keep changes as concise and readable as possible while completing each task.
- Smoke test every code change at least once before presenting it to the user.
- You will go to jail if you provide code that doesn't run.

With AGENTS.md

You: Go evaluate a Meta Model that only uses models built on the latest dataset

I'll start by checking the code in path/to/meta_model/code

I'll reuse the existing Meta Model code and just change the subset of user models

I'll use the path/to/tournament/queries to understand how to filter users by data version

Teach your AI the specifics

Without Teaching Specifics

- AI will reinvent the wheel over and over, not reusing code reliably
- AI will do the same task slightly differently every time
- **Won't retain working knowledge from session to session**

SKILL.md

1. Defines when to use the skill
2. Gives instructions for the using skill
3. Directs to scripts that will aid in performing this skill

SKILL.md

name: oncall

description: Investigate and respond to Numerai oncall alerts

Oncall Response

Quick start

- Capture alert metadata (dag/task, run date, environment, link, error snippet).
- Map the alert to the owning area (airflow2 scheduler vs data_gen pipeline vs data quality validator).
- Classify the alert: data-quality validator, late/missing data, pipeline failure, or infra.
- Run the targeted check for the date(s) in question.
- Produce a concise investigation report with findings and next steps.

SKILL.md

✓  oncall

✓  scripts

 __init__.py

 compositefigi_change_report.py

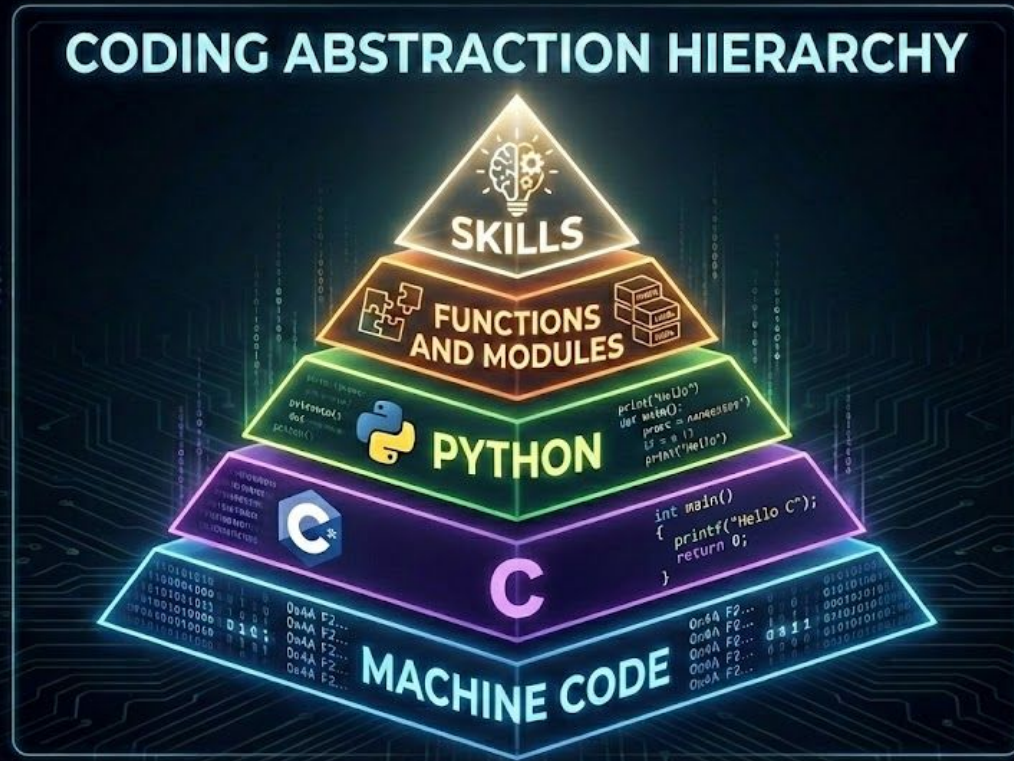
 pagerduty_incident_note.py

 quality_validation_outliers.py

 __init__.py

 SKILL.md

This is the next layer of abstraction



Skills let you just teach AI a task *one time*

All subsequent AIs inherit that learned ability

Example Skill #1

- Design experiment
 - change one variable at a time
 - start with small models to iterate quickly and then scale later
 - compare to existing SotA

Example Skill #2

- Speed up run-time
 - start by making three test cases to ensure no logical change
 - profile the code to identify most valuable parts to speed up
 - make changes one at a time and validate speed improvements and no behavioral changes

Example Skill #3

- Evaluate Models
 - run a particular set of metrics on each set of predictions
 - check stats to see which is better
 - care primarily about BMC but also some about CORR

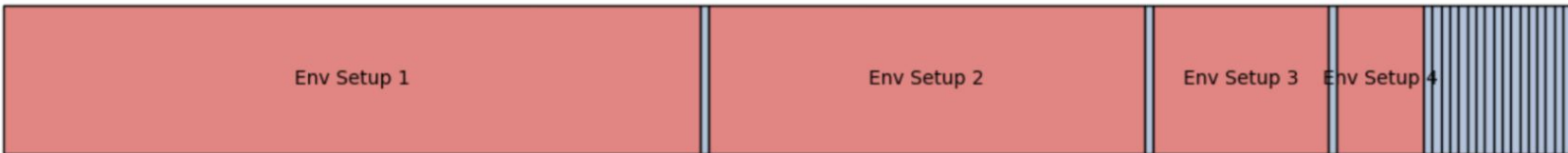
You can compose skills together

- numerai-research skill
 - First use the design-experiment skill
 - Then use the run-model-on-aws skill for all individual models in the experiment design
 - Then use the evaluate-model skill on all resulting models

You can compose skills together

- numerai-research skill
 - First use the design-experiment skill
 - Then use the run-model-on-aws skill for all individual models in the experiment design
 - Then use the evaluate-model skill on all resulting models

AI Automation



All you have to remember



1. Give them access

Codex CLI /
Claude Code CLI



2. Give them context

AGENTS.md



3. Teach them the specifics

SKILL.md



Thanks



Thanks

